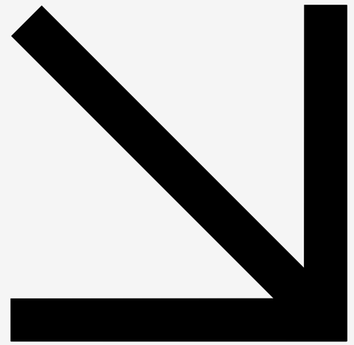
ENTERPRISE FINTECH DATA PLATFORM (EFTP)

Building a Unified Digital Payments & Financial Transactions Analytics Platform
UC12 — Digital Payments & Financial Transactions Analytics

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PROBLEM STATEMENT

Business Problem

ABC FinTech Solutions Ltd. receives financial transaction data from multiple independent enterprise systems.

These systems generate data in different formats, making it difficult to:

- integrate financial data
- maintain consistency
- monitor transactions
- identify fraud
- analyse customer behaviour
- track merchant activity
- support financial reporting

Our project aims to establish a unified data platform that can collect and organize data from these heterogeneous sources.



PROJECT OBJECTIVES

Objectives

Our platform aims to:

- Integrate data from multiple FinTech source systems
- Support heterogeneous data formats
- Establish a structured data ingestion process
- Create a centralized raw-data landing/staging area
- Maintain data definitions and relationships
- Prepare the foundation for future analytics and reporting

The project specifically includes sources such as UPI, digital wallets, payment gateways, merchant systems, CRM, KYC, settlement, mobile banking, fraud detection and API gateway logs.



PROJECT SCOPE

DA1 Scope
Our current submission covers:

Sprint 0

- Project initiation
- Business requirements
- Architecture
- Project planning
- Repository setup

Sprint 1

- Data source identification
- Synthetic dataset generation
- Data dictionary
- Data ingestion
- PostgreSQL staging

ENTERPRISE DATA SOURCES

Source Systems

We designed synthetic data representing 10 enterprise sources:

Source System - Format

- UPI Transaction System - CSV
- Digital Wallet Platform - CSV
- Payment Gateway - JSON
- Merchant Portal - Excel
- CRM - CSV
- KYC Verification System - JSON
- Settlement Processing System - CSV
- Mobile Banking Application - JSON
- Fraud Detection System - CSV
- API Gateway Logs - XML

- These source systems and formats are documented in our Data Dictionary.

DATA DICTIONARY

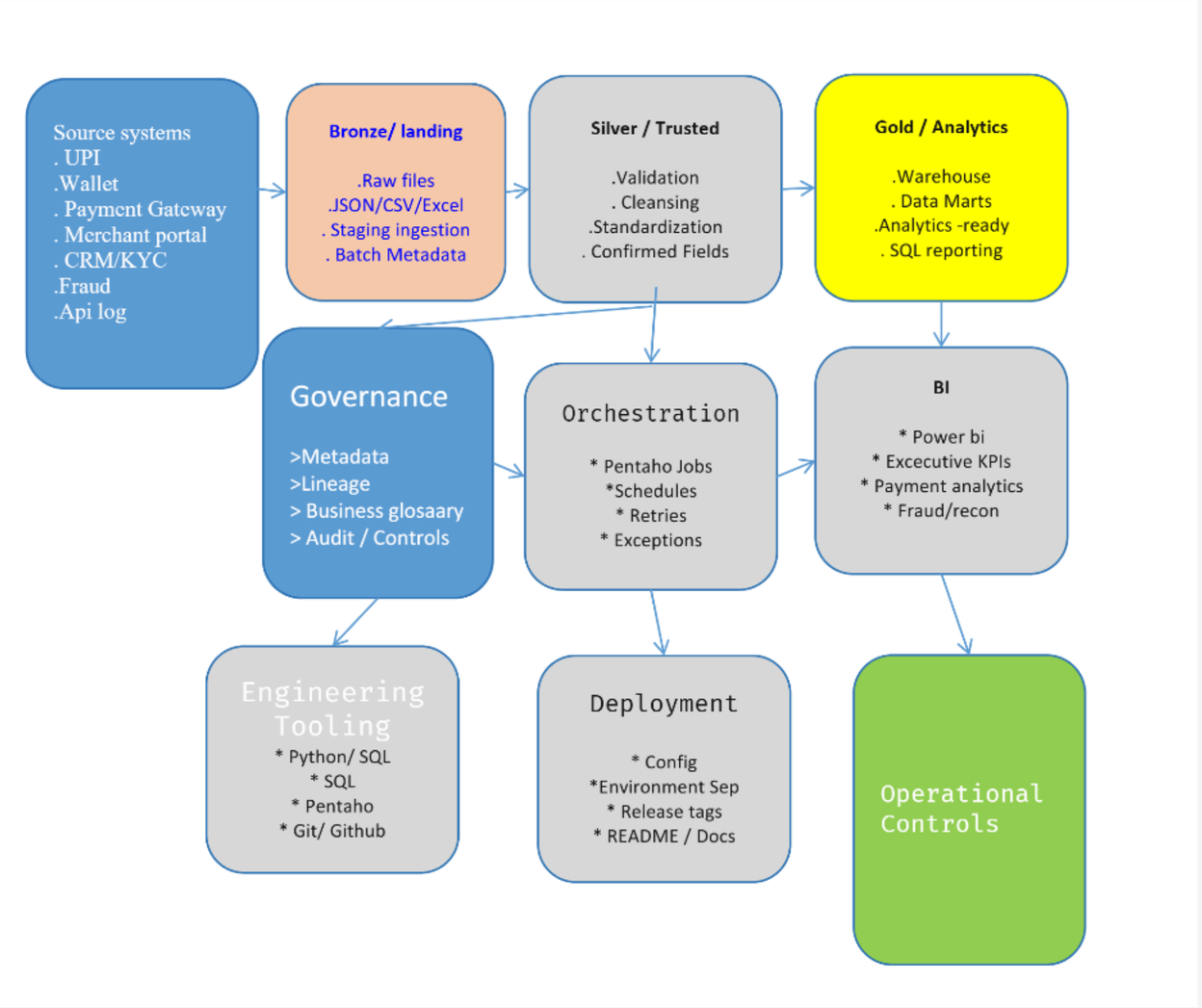
Data Dictionary & Join Keys

The Data Dictionary documents:

- source system
- file format
- field names
- data types
- field descriptions
- primary/foreign keys
- relationships between datasets

Field	Type	Purpose
transaction_id	Text	Transaction identifier
customer_id	Text	Customer identifier
merchant_id	Text	Merchant identifier
amount	Float	Transaction amount
status	Text	Transaction status

SYSTEM ARCHITECTURE



DATA INGESTION

Data Ingestion using Pentaho

We use Pentaho Data Integration to ingest data from heterogeneous sources.

Supported inputs:

CSV

Excel

JSON

XML

SQL / Database

General flow:

Source File

↓

Pentaho Input

↓

Basic Transformation

↓

PostgreSQL

↓

Staging Table

POSTGRESQL STAGING

Staging Layer

The raw data is loaded into PostgreSQL staging tables.

```
PostgreSQL
│
├── stg_upi_transactions
├── stg_wallet_transactions
├── stg_payment_gateway
├── stg_merchants
├── stg_customers
├── stg_kyc
├── stg_settlements
├── stg_mobile_sessions
├── stg_fraud_flags
└── stg_api_logs
```

Purpose

The staging layer provides a centralized location for the ingested raw data before the cleansing and transformation stages.

DATA QUALITY CONSIDERATIONS

Intentional Data Quality Issues

Our synthetic datasets contain controlled data-quality issues so that they can be addressed in later processing.

Examples include:

- Duplicate transaction records
- Null and negative transaction amounts
- Inconsistent status casing
- Inconsistent merchant naming
- Missing customer records
- Settlement amount discrepancies

These issues are intentionally present in the synthetic dataset and are documented for later profiling and cleansing.

GITHUB REPOSITORY

Version Control & Collaboration

We use GitHub to manage the project and collaborate as a team.

Our repository contains:

docs/

datasets/

bronze/

silver/

gold/

pentaho/

python/

sql/

metadata/

lineage/

dashboards/

tests/

CURRENT OUTCOME

What We Achieved

At the end of DA1, we have established:

10 Enterprise Sources

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Multiple Data Formats

↓

Synthetic Financial Dataset

↓

Data Dictionary & Join Keys

↓

Pentaho Ingestion

↓

PostgreSQL Staging

SPRINT 2 — DATA PROFILING AND QUALITY ASSESSMENT

Objective: Identify data-quality issues in the 10 enterprise source tables before analytical processing.

Activities performed:

Profiled all 10 source tables to understand their structure, completeness and consistency.

Identified duplicate transaction IDs and missing or invalid transaction amounts.

Detected inconsistent merchant naming and transaction status formats.

Identified customer IDs referenced in transactions but missing from CRM.

Generated data profiling reports and documented data-quality findings.

Outcome: Established a measurable baseline of data quality for the cleansing and transformation stages.

DATA CLEANING — SILVER LAYER

The Silver layer contains cleaned and standardized data derived from the PostgreSQL staging tables.

Transformations performed:

Removed duplicate transaction records within each transaction source.

Converted transaction amounts into numeric values and excluded invalid or non-positive amounts.

Standardized transaction status values to uppercase.

Standardized merchant names for consistent representation.

Preserved source records and relationships needed for downstream processing.

Technology: PostgreSQL, Python and Pentaho Data Integration.

Outcome: Created 10 cleaned Silver tables ready for analytical warehouse loading.

DATA QUALITY FINDINGS

Data Source	Duplicate IDs	Invalid / Missing Amounts	Non-Positive Amounts
UPI Transactions	69	37	21
Digital Wallet Transactions	32	13	10
Payment Gateway Transactions	19	11	5

- 20 distinct customer IDs referenced in transactions were missing from the CRM customer records.
- Merchant naming inconsistencies were identified across the source data.
- Some transactions lacked customer or merchant identifiers.

Documented data-quality issues and exceptions in the Data Quality Report.

STAR SCHEMA — GOLD DATA WAREHOUSE

The Gold layer organizes cleaned financial data into fact and dimension tables to support reporting and analytics.

Dimension tables

gold.dim_date — Date, month, quarter, year and day information.

gold.dim_channel — Payment channel details. (Digital , Wallet and Gateway)

gold.dim_customer — Customer attributes and KYC information.

gold.dim_merchant — Merchant details and settlement attributes.

Fact tables

gold.fact_transactions — Transaction amount, status, payment channel, customer, merchant and fraud-related information.

gold.fact_settlement — Settlement counts, transaction values, settled amounts and settlement status.

Outcome: Established a centralized analytical model for transaction and settlement reporting.

ETL WORKFLOW — SILVER TO GOLD

Data flow:

PostgreSQL Staging

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Data Profiling and Cleansing (using Python)

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Silver Layer(storing cleansed data to silver layer schema)

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Pentaho Transformations

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Gold Dimensions and Fact Tables

↓

SQL Reporting and Analytics

Created six Pentaho transformations for loading the Gold dimensions and fact tables.

Used SQL-based transformations to select, join and standardize data.

Implemented incremental loading logic to avoid inserting existing records again.

Mapped source attributes to the corresponding warehouse columns.

Outcome: Populated the Gold warehouse with cleaned transaction and settlement data.

The Gold warehouse was validated using SQL queries.

Table	Validated Row Count
`gold.dim_date`	180
`gold.dim_channel`	3
`gold.dim_customer`	1,980
`gold.dim_merchant`	150
`gold.fact_transactions`	11,905
`gold.fact_settlement`	3,864

- No duplicate transaction IDs in the transaction fact table.
- No missing transaction date keys, channel keys or amounts.
- No missing settlement date or merchant keys.
- Confirmed completeness of eligible records loaded into Gold.

Outcome:The warehouse passed the main structural and completeness checks.

REPORTING USING SQL

SQL queries were developed to demonstrate how the Gold warehouse supports business analysis.
Reports implemented:
Payment Channel Success Rate — Compares transaction success rates across UPI, digital wallets and payment gateways.
Top Merchants by Transaction Value — Identifies merchants with the highest total transaction value.
Daily Fraud Rate — Analyses the proportion of transactions flagged as fraudulent by date.
Sample results:
Digital Wallet success rate: 87.72%
Payment Gateway success rate: 87.09%
UPI success rate: 87.43%

Outcome: Enabled business-oriented analysis of payment performance, merchant activity and fraud patterns.

SPRINT 2 — KEY ACHIEVEMENTS

Profiled 10 enterprise source tables.

Generated data profiling and data quality reports.

Created 10 cleaned Silver tables.

Designed and populated the Gold star schema.

Loaded 4 dimension tables and 2 fact tables.

Validated warehouse records using SQL.

Developed reporting queries for payment performance, merchant transaction value and fraud analysis.

Documented source-data limitations and unresolved identifier gaps.

Overall outcome: Progressed from raw financial data in PostgreSQL staging to a cleaned, structured and validated analytical data warehouse.

Thank You